

HDOC Day-24

Increasing Patterns Using Switch Statement and Nested Loops

1. Increasing right-angled triangle

Algorithm

1. Start.
2. Input number of rows n.
3. For i = 1 to n, print '*' i times using an inner loop.
4. Move to the next line after every row.
5. Stop.

Sample output

```
n = 4
*
* *
* * *
* * * *
```

2. Increasing row-number triangle

Algorithm

1. Start.
2. Input number of rows n.
3. For i = 1 to n, print the row number i exactly i times.
4. Move to the next line after every row.
5. Stop.

Sample output

```
n = 4
1
2 2
3 3 3
4 4 4 4
```

3. Increasing natural-number pattern

Algorithm

1. Start.
2. Input number of rows n.
3. For each row i, print numbers from 1 to i.
4. Move to the next line.
5. Stop.

Sample output

```
n = 4
1
1 2
1 2 3
1 2 3 4
```

4. Consecutive-number triangle

Algorithm

1. Start.
2. Input number of rows n and initialize num = 1.
3. For each row i, print i consecutive values of num.
4. Increment num after every printed value.
5. Stop.

Sample output

```
n = 4
1
2 3
4 5 6
7 8 9 10
```

5. Increasing alphabet pattern

Algorithm

1. Start.

2. Input number of rows n ($1 \leq n \leq 26$).
3. For each row i , print uppercase letters from A through the i -th letter.
4. Move to the next line.
5. Stop.

Sample output

```
n = 4
A
A B
A B C
A B C D
```

C Program

```
#include <stdio.h>

int main() {
    int choice, n, i, j, num = 1;

    printf("INCREASING PATTERNS MENU\n");
    printf("1. Increasing right-angled triangle\n");
    printf("2. Increasing row-number triangle\n");
    printf("3. Increasing natural-number pattern\n");
    printf("4. Consecutive-number triangle\n");
    printf("5. Increasing alphabet pattern\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    printf("Enter number of rows: ");
    scanf("%d", &n);

    switch (choice) {
        case 1:
            for (i = 1; i <= n; i++) {
                for (j = 1; j <= i; j++) printf("* ");
                printf("\n");
            }
            break;

        case 2:
            for (i = 1; i <= n; i++) {
                for (j = 1; j <= i; j++) printf("%d ", i);
                printf("\n");
            }
            break;

        case 3:
            for (i = 1; i <= n; i++) {
                for (j = 1; j <= i; j++) printf("%d ", j);
                printf("\n");
            }
            break;

        case 4:
            num = 1;
            for (i = 1; i <= n; i++) {
                for (j = 1; j <= i; j++) printf("%d ", num++);
                printf("\n");
            }
            break;

        case 5:
            if (n < 1 || n > 26) {
                printf("Enter rows between 1 and 26.\n");
                break;
            }
            for (i = 1; i <= n; i++) {
                for (j = 0; j < i; j++) printf("%c ", 'A' + j);
                printf("\n");
            }
            break;

        default:
            printf("Invalid choice.\n");
    }
    return 0;
}
```

Test Cases

Test	Choice	n	Expected result
1	1	4	4-row increasing asterisk triangle
2	2	4	Rows: 1; 2 2; 3 3 3; 4 4 4 4
3	3	4	Natural numbers from 1 to row number
4	4	4	Consecutive numbers 1 through 10 in triangle
5	5	4	A; A B; A B C; A B C D
6	5	27	Error: rows must be between 1 and 26
7	9	4	Invalid choice

Result: The C program was written using a switch statement and nested loops, and the five increasing patterns were generated successfully for valid inputs.